

1. Project Title

GeoAgriSense:
Spatial Intelligence and Multimodal AI for Agricultural Field Scouting

2. Problem Statement

Agricultural field scouting requires farmers and field workers to identify crop problems, understand where they are occurring and determine which issues require immediate attention.

Field observations may contain useful information such as crop symptoms, severity and location, while photographs provide additional visual evidence.

GeoAgriSense explores the integration of these information sources with multimodal artificial intelligence.

3. Target Users

Primary users:

- Smallholder farmers
- Commercial farmers
- Agricultural field scouts
- Farm managers
- Agricultural extension personnel

Secondary users:

- Agronomists
- Agricultural organisations
- Agricultural researchers

4. Proposed Solution

GeoAgriSense is a spatial agricultural decision-support prototype.

It combines:

- geospatial farm information;
- production blocks;
- field observations;
- scouting information;
- crop photographs; and
- Gemini multimodal AI.

The objective is to help users move from:

"Something appears wrong with my crop"

towards:

"Where is the problem, what was observed, what does the photograph show, and what should I investigate next?"

5. System Workflow

The primary workflow is:

User
↓
Select field observation
↓
Upload crop photograph
↓
Application retrieves structured observation context
↓
Application combines image + field context
↓
Gemini API
↓
Multimodal analysis
↓
Structured scouting assessment
↓
User reviews recommendation
↓
Human verification / field action

6. System Architecture

The system consists of the following components:

Presentation Layer
Streamlit dashboard and interactive map.

Data Layer
GeoJSON and CSV demonstration datasets.

Geospatial Layer
Folium / GeoJSON visualisation.

Application Logic
Python services for observation processing and Gemini interaction.

AI Layer
Google Gemini API.

Configuration Layer
Environment variables containing API configuration.

7. Technologies Used

Python
Streamlit
Folium
Streamlit-Folium
Pandas
GeoPandas
GeoJSON
Google Gemini API
Google GenAI Python SDK
python-dotenv

8. Gemini API Integration

Gemini is a functional component of the application.

The multimodal workflow accepts a crop photograph and structured observation context.

The structured context may include:

- crop
- block
- symptom
- severity
- field notes
- observation date
- geographic coordinates

The application sends the photograph and textual context to the Gemini model.

Gemini analyses both information sources.

The returned response is displayed as a scouting report containing:

- visual observations
- crop condition
- possible causes
- risk assessment
- recommended field checks

- recommended next action
- limitations

9. Why Gemini Is Important

The value of Gemini is not simply that it generates text.

Its multimodal capability allows GeoAgriSense to combine visual crop evidence with structured field information.

This enables a richer interaction than a text-only agricultural chatbot.

The geospatial component answers:

"Where is the problem?"

The field observation answers:

"What did the scout observe?"

The photograph provides:

"What does the crop look like?"

Gemini provides:

"What might this visual and contextual evidence indicate, and what should be checked next?"

10. Development Process

The team began by identifying agricultural field scouting as a meaningful problem area.

The initial concept was expanded to incorporate geospatial intelligence because generic agricultural advisory applications are common hackathon concepts.

The team therefore focused on connecting agricultural observations to geographic locations.

During implementation, the project was developed incrementally:

1. Environment setup
2. GitHub collaboration
3. Streamlit application
4. Geospatial data
5. Farm map
6. Field observations
7. Gemini API authentication
8. Gemini text interaction

9. Multimodal crop analysis

10. Dashboard integration

11. Technical Challenges

Challenges included:

- development environment configuration;
- collaborative Git workflow;
- geospatial package installation;
- Gemini model/API availability;
- temporary Gemini API service availability;
- integrating existing team data into the application;
- developing within the limited hackathon timeframe.

12. Testing

The team tested:

- application startup;
- data loading;
- map rendering;
- field observation display;
- Gemini API authentication;
- text generation;
- image upload;
- multimodal crop analysis.

The final prototype should be demonstrated live during judging.

13. Limitations

The prototype is not a production agricultural diagnostic platform.

Limitations include:

- demonstration datasets;
- dependence on network connectivity;
- potential AI hallucination;
- image-quality dependence;
- limited crop/disease knowledge validation;
- absence of comprehensive sensor integration;
- limited historical data;
- no formal agronomist validation layer.

14. Responsible AI / Risk Considerations

The system deliberately frames Gemini output as decision support.

Potential risks include:

Incorrect recommendations

AI may misinterpret visual symptoms.

Over-reliance

A farmer may treat an AI output as a definitive diagnosis.

Data privacy

Farm locations and agricultural information may be sensitive.

Security

API credentials must be protected.

Human oversight

Important agricultural decisions should remain subject to human verification.

15. Future Improvements

With additional development time, the team would add:

- IoT sensors;
- weather APIs;
- satellite imagery;
- vegetation indices;
- historical disease datasets;
- automated spatial risk maps;
- temporal crop monitoring;
- mobile/offline scouting;
- agronomist validation;
- automated alerts;
- predictive analytics.

16. Team Contributions

Member Role / Contribution

Butholethu Mthokozisi Dube - Team Lead / Application & Gemini Integration / Testing

Rufaro Nyakudya - Pitch / Documentation

Thembelihle Michelle Ndebele - Product Researcher

Shaun Jeranyama - Data / GIS Assistant

Tichaona B. Wutete - UI/UX + Visual Design

17. Conclusion

GeoAgriSense demonstrates how spatial intelligence and multimodal AI can be combined to improve agricultural field scouting.

The prototype does not attempt to replace farmers or agricultural experts.

Instead, it demonstrates a workflow in which field observations, geographic context and crop imagery are brought together to help users prioritise investigation and make more informed decisions.